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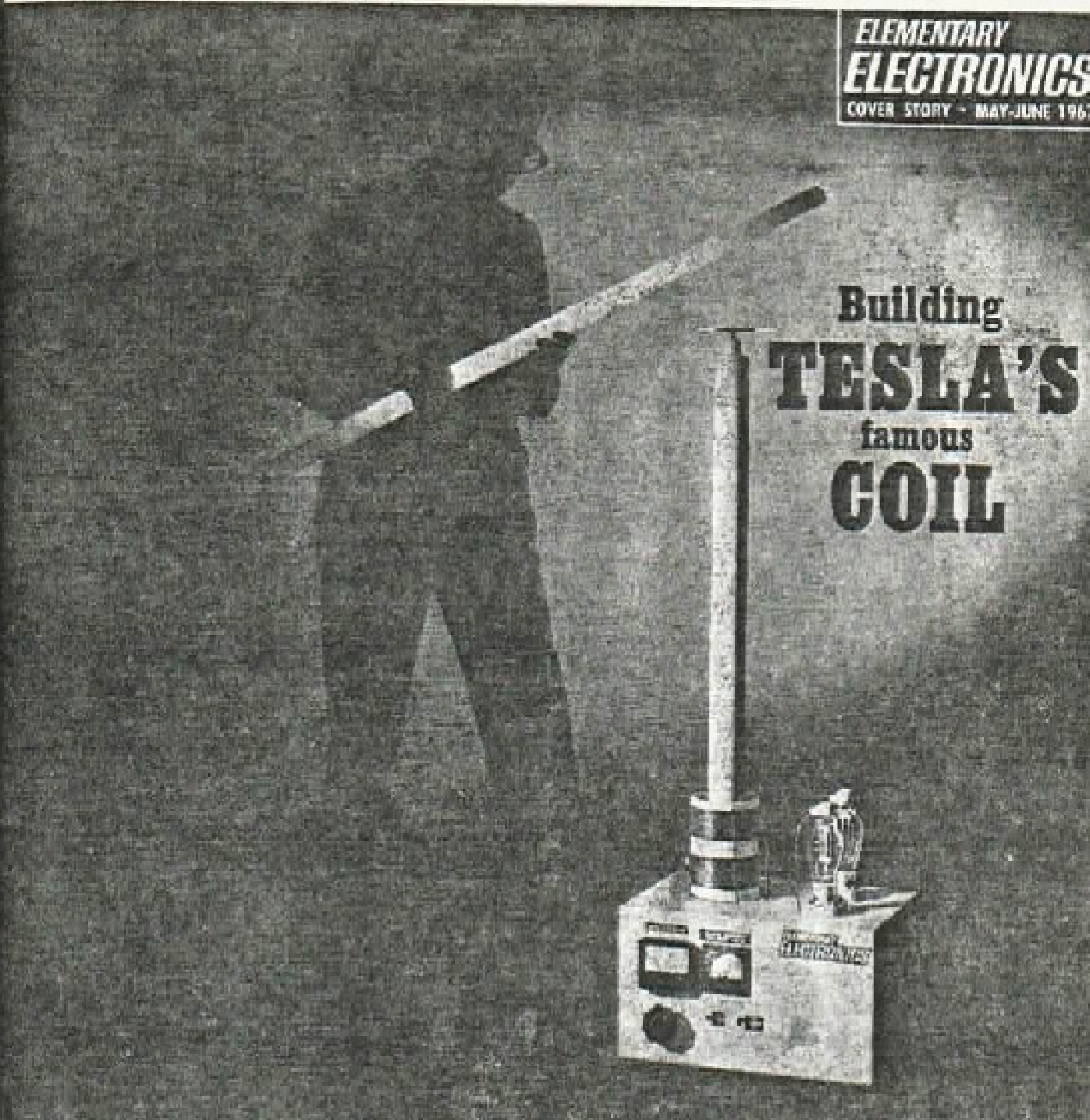


Building TESLA'S famous COIL

By Kenneth A. Wright and Wallace Kern

Science teachers, amateur scientists, magicians, and many electronics hobbyists will find ELEMENTARY ELECTRONICS' *Tesla Coil* an interesting project both to build and to perform experiments with. Readily demonstrating the principles of radio and electricity, the "giant coil" may also lead to an interest in Dr. Tesla, whose work is little known by most beginning science students. Few, in fact, have heard of Nikola Tesla, or of the work he has done. But many experts believe Tesla was as great as Edison—or even greater—in his contributions to science.

Tesla's Contributions. During Dr. Tesla's time, in the late 19th century, only direct current (DC) was used commercially. (turn page)

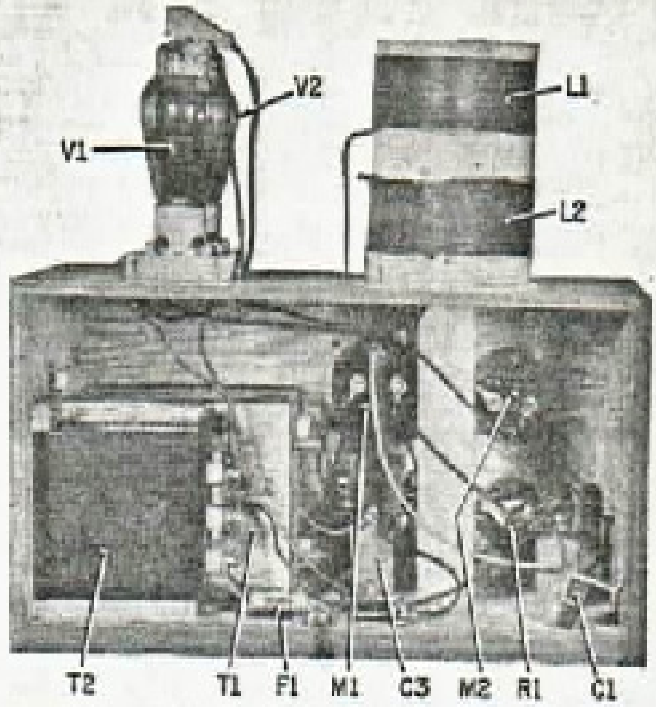
though there was some knowledge of alternating current (AC). Tesla was the first person who investigated the practical uses of alternating currents to any great extent.

A few years after he came to the United States (1884), Tesla worked out several possible systems of alternating-current machinery, dynamos, motors, transformers, and distribution systems for which he was granted 45 patents between 1887 and 1893. In 1891, George Westinghouse purchased some of Tesla's systems and used them at the 1892 Chicago World's Fair to supply all the electrical power needed.

Dr. Tesla proposed a 7500-kilowatt transmitting plant producing RF currents at 100-million volts. One of the chief uses for this transmitted energy would be to illuminate "isolated" homes. The energy would be collected by a terminal a little above the roof of each house.

He actually built a generator similar to the one he proposed and lit 300 electric bulbs 25 miles from his laboratory. He started to build a transmitting station—probably the world's largest Tesla Coil—on a site near New York City before World War I, but ran out of funds.

A Tesla Coil. This high-voltage RF generator is more commonly known as a Tesla Coil, although the actual construction hardly resembles the original apparatus of Dr. Tesla. The Tesla Coil is a dynamic demonstrator of electrical principles, and a crowd pleaser as a science fair project. It can be built for much less than the estimated cost—

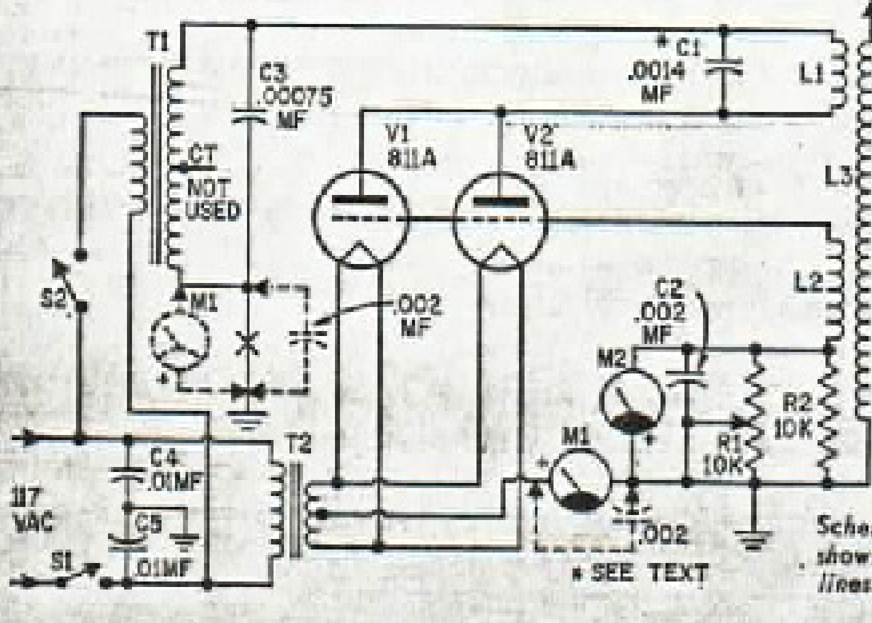


Rear view of this Tesla's Coil shows the location of the major components. Layout isn't critical but all leads to coil, vacuum tubes and bypass capacitors should be kept as short as possible—as in RF wiring.

if you can shop around the "bargain counters" in parts stores. All the materials used in building the unit pictured here were from war-surplus components. (The parts list gives stock, over-the-counter parts. Admittedly, new parts are expensive, so some wise shopping may be in order.) The Tesla Coil is very easily built and it gives good results. Sparks several inches long may be obtained from the coil at voltages on the order of 50,000 volts.

Circuitry. The apparatus consists basically of a plate-tuned oscillator, as can be seen in the schematic diagram. Practically any transmitter-type triode vacuum-tube will serve in the oscillator.

One of the things considered in selecting a tube was that it have a plate voltage rating of about 1500 volts. The higher the voltage, the more spectacular the experiments that may be performed with the apparatus. The second factor considered was that it have a fairly-high power output (of about 150 watts or more). Two 811A transmitting triodes met the specifications. Both of them are used, in



Schematic diagram of the Tesla Coil shows alternate wiring connections in dotted lines. Circuit is quite flexible.

PARTS LISTS FOR TESLA COIL

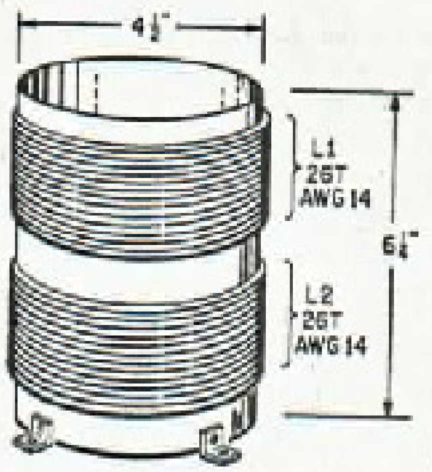
C1—.0014-mf., 3000-volt DC, ceramic-disc or mica capacitor.
(two or more capacitors in parallel—see text)
.0004-mf., (400-mmf.) Centralab DD30-401 or equiv.
.001-mf., (1,000-mmf.) Centralab DD30-102 or equiv.
C2—.002-mf., (2,000-mmf.) 3000-volt DC, ceramic disc capacitor (Centralab DD30-202 or equiv.)
C3—.00075-mf., (750-mmf.) 3000-volt DC, ceramic disc capacitor
C4, C5—.01-mf., 600-volt DC, molded tubular capacitor
F1—6-ampere, glass cartridge fuse
L1—26-turns, AWG-14, enameled wire (see text)
L2—26-turns, AWG-14, enameled wire (see text)
L3—2100-turns, AWG-26, enameled wire (see text)
M1—0-300 milliamperes DC, (Shurite 550, 850, 950 or equiv.—see text)
M2—0-150 volts DC, (Lafayette 99R5040 or equiv.—see text)
R1—10,000-ohm, 50-watt, wirewound potentiometer
(Ohmite Type J-0332 vitreous-enameled rheostat)
R2—10,000-ohm, 10-watt, wirewound resistor
S1, S2—S.p.s.t. toggle switch
T1—Plate transformer, 1840-volt, 250-ma secondary (Allied-Knight 63U700 or equiv.)
T2—Filament transformer, 6.3-volt, 10-ampere secondary (Allied-Knight 61Z418 or equiv.)
V1, V2—Vacuum tube, type 811A
2—4-pin, low-loss, ceramic tube socket (Amphenol 49-RS54, Johnson 123-209-1, or better)
1—Lead-in bushing (E. F. Johnson 135-53 or equiv.)
Misc.—Magnet wire, hookup wire, coil forms, wooden cabinet, wood screws, machine screws, solder, etc.
Estimated cost: $50.00
Estimated construction time: 8 hours

parallel, to deliver a total output of about 300 watts of RF power.

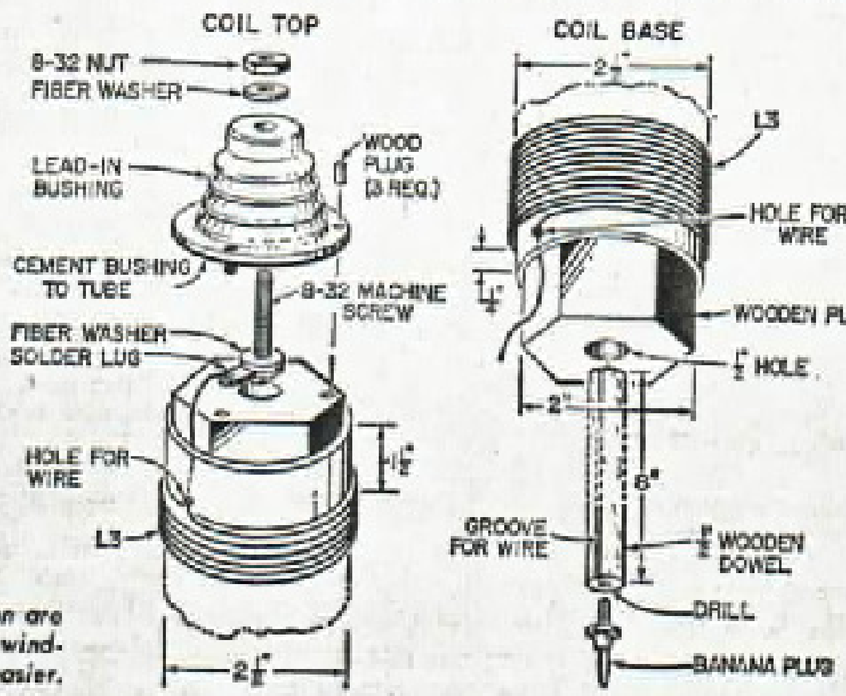
Supply Transformers. A high-voltage-secondary transformer (delivering about 1840 volts at about 250 milliamperes) is used as the AC plate-voltage supply. In the schematic diagram, a 0.00075-μf by-pass capacitor is placed across the high-voltage secondary to prevent the high-frequency current from going through the transformer and out the AC power line.

The filament-transformer secondary supplies 6.3 volts, and 4 amperes are needed for each tube. Since the cathodes in these tubes are directly heated, the current from the plates must flow through the filament transformer to the filament. This is best accomplished by connecting the B— to the centertap of the filament winding, making a complete circuit as shown.

Coil Wiring. Plate coil L1 is 26 turns of AWG-14 enameled magnet wire on a thoroughly wax-impregnated, 4¼-in. diameter cardboard form (phenolic or plastic can also be used). The grid coil (L2) is wound above L1, on



Grid and plate coils are wound with same size wire. Large diameter wire increases "Q" of grid inductance and that of plate tank L1-C1. Low resistance coils have less internal losses and higher output—brush discharge is greater because of better efficiency.



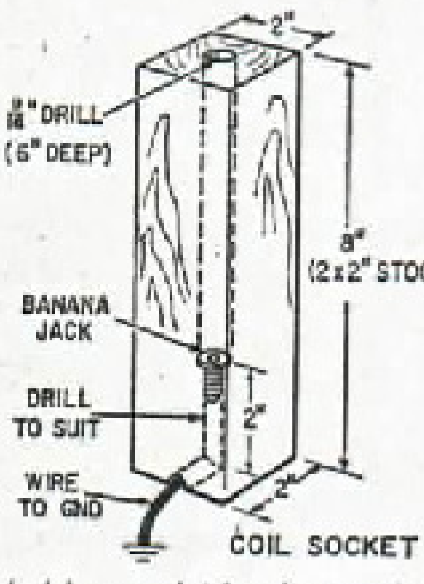
Details of coil construction are important to simplify fabrication and winding. Plug-in coil makes storage easier.

e/e TESLA'S COIL

the same form, in the same direction, and with the same number of turns of AWG-14 wire as L1. Both coils must be given two coats of good insulating varnish. Brackets are put on the bottom of the form to mount it on the wooden cabinet.

To complete the resonant circuit (L1-C1), the value of C1 was calculated so the circuit oscillated at about 500 kiloHertz (kilocycles). From experimentation it was found that the circuit used here oscillates best with a capacitance of about 0.0014-μf (.001-μf and .004-μf in parallel) with a corresponding frequency of about 500 kHz.

The output coil (L3) consists of 2100 closely-wound turns of AWG-26 enameled wire on a thoroughly wax-impregnated 2¼-in. diameter cardboard form. Three coats of insulating varnish were put on the windings. One end of the winding of the coil is connected to the terminal of a cone-shaped standoff insulator. A wooden plug is fitted



Wood block forms socket for plug-in coil L3. Six-inch deep hole accepts dowel which takes strain off jack and plug. Auger bits have necessary length for hole.

to the inside of the bottom portion of the coil with a dowel inserted in a hole drilled in the plug to mount the coil and make the ground-end connection. Shellac the wood to keep out moisture.

A box, with the back open, should then be constructed. Both triode tubes are mounted on top of the box along with L3—mounted concentrically with L1 and L2. Both transformers (T1 and T2) are mounted inside the box at one side.

AC Input. Two 0.01-μf capacitors, connected from each side of the AC-line input to ground, prevent any high-frequency oscillations from escaping to the power lines. The transformers are protected with a 6-ampere fuse as shown in the schematic diagram—a circuit breaker can be used.

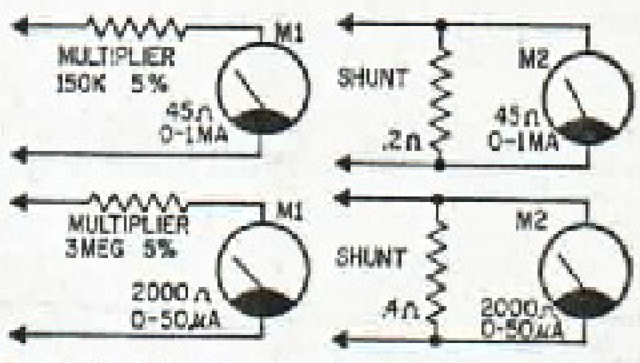
Power Switching. Two switches are mounted on the front panel of the box. Plate transformer (T1) can not be turned *on* with S2 unless the filament transformer (T2) is *on*. This protects the tubes. For your own safety, it is wise to ground the apparatus to a waterpipe on some other ground.

In order to set R1 correctly, a DC voltmeter (M2) is wired across it to measure the grid bias. A DC milliammeter (M1) is also connected between the center tap of the filament transformer and ground to indicate the total plate and grid currents.

With the circuit completely wired—and double-checked for wiring mistakes—S1 may be turned *on* for filament warmup. After approximately 30 seconds, the plate-power transformer T1) is turned *on* and the apparatus is in operation.

Control R1 is adjusted for a grid bias of about 115 volts—the plate-current meter should read about 200 milliamperes. With the proper control adjustments some sort of brush discharge from the insulator terminal should be clearly visible.

With a variety of capacitors on hand, C1 can be changed in total capacitance until the brush discharge is maximum. With this condition maximum oscillation is obtained. (Maximum oscillation will vary somewhat with coil construction.) It was noticed that



Even 0-1 milliampere or 0-50 microampere meters may be used to measure plate current and grid voltage in the Tesla Coil circuit if proper multiplier or shunt is added to convert meter to indicate proper range.

whenever the incorrect values of either R1 or C1 are used, the brush discharge was not as great.

The plates of the tubes must be watched at all times to make certain that their color does not get brighter than their normal dull-red. (*Continued on page 102*)

The C-Battery

Continued from page 41

tive-going half-cycles drives the grid positive. This causes the grid to attract a small portion of the electron stream traveling from cathode to plate. These electrons charge the capacitor plate negatively, as shown. As the applied AC signal swings less positive, toward the negative part of the cycle, the capacitor retains some of its negative charge. The only way electrons stored on the capacitor plate can run off is by "leaking" to ground through the grid-leak resistor. Since the resistor slows down the run-off, the capacitor keeps the grid negatively biased throughout the signal cycle.

An advantage of grid-leak bias is that it's self-regulating. Should an oscillator, for example, increases output due to a temporary surge in the power supply, the stronger signal generates higher negative bias. It tends to reduce output to normal. The same action occurs in the regenerative detector of simple receivers; where the increase in grid-leak bias acts as a form of automatic gain control.

A disadvantage of grid-leak bias is that signal failure removes all bias from the tube grid. If the tube is a high-power transmitter type, it could suffer a disastrous increase in plate current. But in many cases, some form of protection is built into the circuit; usually a small amount of fixed or cathode bias, possibly a voltage-regulator tube is used as a fixed-voltage drop.

No Bias. Finally, there's the no-bias system. A special class of high-power tube is designed to operate at a zero-bias level in certain applications, such as single sideband-transmitter circuits. When a signal is applied to the zero-bias tube, its grid draws current uniformly throughout the cycle. The tube appears as a steady load to the previous stage in the transmitter, a factor which helps to reduce distortion. ■

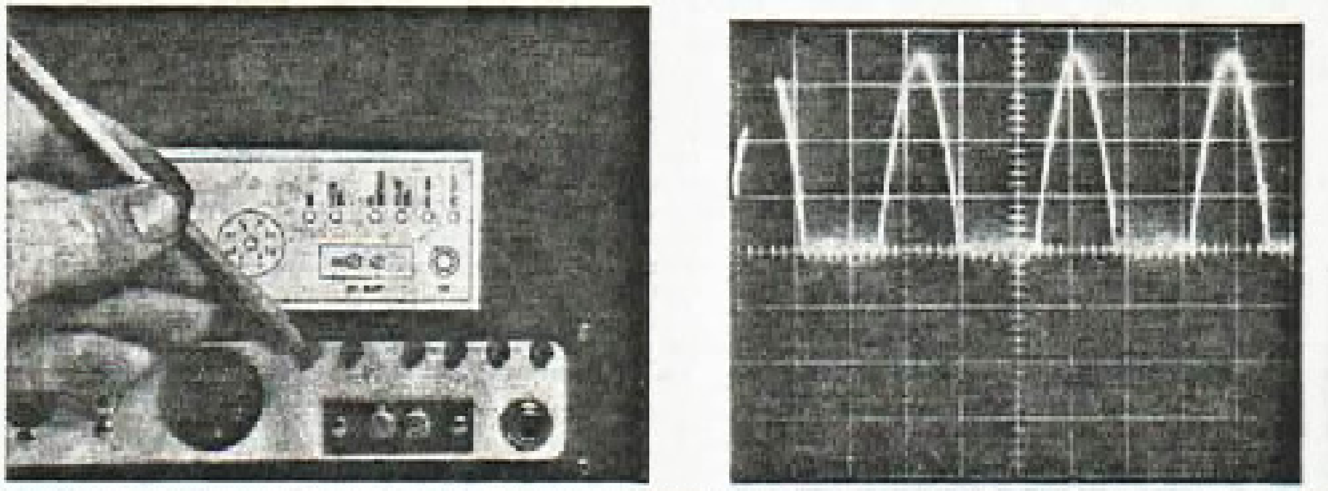


Fig. 11. Adjustments on the rear apron of this transmitter set bias among other things. Sometimes control function is accomplished by a bias adjustment rather than by changing some other circuit parameter. No-bias operation waveform (above, right) results when bias control (left) is properly adjusted on this rig.

Tesla's Coil

Continued from page 30

Because each builder will want to use parts he has available, it is inadvisable to give specific instructions. Even substitutions for meters M1 and M2 may be made by using their proper shunt or multiplier. Though cardboard forms, shellac, and AC-fed oscillators may seem haywire today, the use of more deluxe components and circuitry is not justified for the average experimenter.

Of course you can build a lower-power version of the Tesla coil for a lot less money. But with less power available the experiments will be limited and many may be impossible to perform. Lower power will give less brush discharge.

Experiments with the unit will reveal the many fascinating and spectacular demonstrations which can be performed—such as lighting fluorescent and neon tubes without wire connections. In fact, the energy of the discharge is sufficient to ignite candles and paper, or to drive a pin-wheel balanced on the discharge point. Even the smallest Tesla coil never fails to astound and interest onlookers with feats such as these. ■